

Wormhole Foundation EVM-CCTP Integration Audit Summary

Prepared by Cyfrin

Version 2.1

Lead Auditors

Giovanni Di Siena

0kage

Assisting Auditors

Hans

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Wormhole Foundation on Wormhole Foundation EVM-CCTP Integration. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.1 dated April 9, 2024, was delivered to Wormhole Foundation, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 21 days, the Cyfrin team conducted an audit on the Wormhole Foundation EVM-CCTP Integration code provided by Wormhole Foundation. The findings consist of 2 Medium & 3 Low severity issues with the remainder being informational.

Summary

Project Name	Wormhole Foundation EVM-CCTP Integration
Repository	wormhole-circle-integration
Commit	f7df33b159a7...
Audit Timeline	Jan 15th - Feb 12th, 2024
Methods	Manual Review, Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	2	0	2
Low Risk	3	0	3
Informational	8	2	6
Gas Optimizations	0	0	0
Total	13	2	11

6 Audit Scope

Cyfrin conducted an audit of the Wormhole CCTP integration contracts based on the code present in the repository commit hash f7df33b.

The following directories were included in the scope of the audit:

- evm/src/contracts/*
- evm/src/libraries/*

No existing Wormhole/CCTP smart contracts or their respective off-chain components were included.

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Wormhole is a generic message passing protocol that enables communication between blockchains. CCTP is a permissionless on-chain utility that enables supported Circle assets (USDC/EURC) to move between supported blockchain networks. Together, the Wormhole CCTP integration allows USDC/EURC to be transferred between supported CCTP domains along with a generic Wormhole payload.

Verifiable Action Approvals (VAAs) are the core messaging primitive in Wormhole: Guardian signatures are aggregated and combined into a VAA, which is relayed to the target chain and processed by a receiving contract. To transfer a Circle asset, it is burned on the source chain, a signed attestation is fetched from Circle, and the asset is minted on the destination chain along with the execution of additional logic.

The architecture of the integration primarily consists of the ICircleIntegration contract, composed of the Governance and Logic/WormholeCctpTokenMessenger APIs, which interfaces with the Wormhole Core Bridge and the CCTP TokenMessenger and MessageTransmitter contracts.

9 Attestation

Cyfrin attests that it performed a security review of the Wormhole Foundation EVM-CCTP Integration code provided by Wormhole Foundation.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-04-09-cyfrin-wormhole-evm-cctp-v2-1.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Wormhole Foundation has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=66bafdb9885af2aa9a7ec343982b1a75a6034fc33dd1f1d241bb28229d996ce1
F=2024-04-09-cyfrin-wormhole-evm-cctp-v2-1-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.